SHASHWAT SAHU

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EDUCATION

Bangalore Institute of Technology, Bengaluru — B.E. Robotics & Artificial Intelligence in Robotics & Artificial Intelligence *(2027)* **| CGPA: 2023–2027**

KEY TECHNICAL PROJECTS

PARASYTE — Adaptive Robotic Octopod [Embodied AI, Robot Learning, Embedded Robotics, SNN-RNN, Surprise Detection]

- Architecting an 8-legged embodied AI system for autonomous locomotion and navigation through predictive sensor-error minimization, without engineered gait tables, reward functions, or simulation pre-training.
- Building a hybrid SNN–RNN architecture with adaptive surprise detection, gradient-based credit assignment, prioritized failure replay, and continual-learning mechanisms for self-supervised behavioral adaptation.
- Patent application and research paper in preparation.

SynaptIArm-6X — Neural-Augmented Robotic Manipulator [Physical AI, Embedded ML, EMG/EEG Signal Processing, Inverse Kinematics, Servo Control]

- Developed a 6-DOF manipulator translating EMG/EEG-derived biosignals into robotic actions using real-time signal processing and ML-based intent recognition.
- Achieved 94% intent-detection accuracy with <30 ms inference latency; integrated filtering, feature extraction, servo control, and inverse-kinematics-based manipulation.

Custom 12-DOF Quadruped Robot ("Shashwat's Bot") [Embedded Robotics, Motion Control, Raspberry Pi 4, ESP32, IMU Feedback]

- Designed and built a 12-DOF quadruped integrating mechanical design, embedded electronics, motor control, Raspberry Pi 4, ESP32, IMU-feedback stabilization, and Python-based inverse kinematics.

WORK & LEADERSHIP EXPERIENCE

AI & Robotics Operations Manager Intern — Incanto Dynamics *(2025 – Present)*

- Supported testing, integration, and operational deployment of robotics and AI-driven systems in real-world environments.
- Worked across system validation, deployment, and technical operations; contributed to evaluating and improving the reliability of integrated robotic workflows.

Robotics Intern — Tech Analogy *(2024 – 2025)*

- Worked on robotic automation and software simulation, contributing to the development and evaluation of robotics workflows.
- Gained hands-on exposure to ROS-based system integration and robotics software environments.

Senior Product Manager — AIESEC in India *(2024 – 2025)*

- Engineered and optimized acquisition and conversion workflows, generating ^{1}1.6L in direct value and contributing toward a ^{1}5.4L entity-wide milestone.
- Improved funnel efficiency by 35% and conversion performance by 40% through experimentation, workflow optimization, and data-driven iteration; led a 5-member cross-functional team.

Artificial Intelligence Intern — 1Stop.ai *(Apr 2024 – Jun 2024)*

- Worked on ML models and supported API-based AI integration, gaining exposure across model experimentation and deployment-oriented workflows.

Club Operations Lead & Technical Team Manager — Robotics Cell, BIT *(Jan 2024 – Jan 2025)*

- Led student robotics initiatives across physical robot development, embedded systems, technical execution, and hackathon delivery; coordinated mechanical, electronics, and software teams.

TECHNICAL SKILLS & COMPETENCIES

Programming: Python, C++, C, JavaScript
Robotics & Physical AI: ROS2, Arduino, Raspberry Pi, Gazebo, Fusion 360
AI & ML: PyTorch, TensorFlow, OpenCV, Scikit-learn
Tools & Frameworks: Git, FastAPI, React.js, Supabase